Suraj Sah Kanu

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Professional Summary

Machine Learning Engineer and AI Researcher with a proven track record of designing, developing, and deploying enterprise-ready AI systems. Expert in architecting advanced **Retrieval-Augmented Generation (RAG)** pipelines, **Multi-Agent AI Systems** (**LangGraph, CrewAI**), and custom LLM fine-tuning workflows utilizing parameter-efficient methods (**QLoRA/LoRA**). Adept at bridging the gap between cutting-edge AI research and production environments by building scalable APIs with **FastAPI**, containerizing applications via **Docker**, and implementing robust MLOps practices on cloud platforms.

Technical Skills

Programming Languages: Python, SQL, Bash

Machine Learning & Deep Learning: PyTorch, TensorFlow, Scikit-learn, YOLO, Transfer Learning, Computer Vision

LLM & Generative AI: Hugging Face Transformers, QLoRA, LoRA, PEFT, TRL, Unsloth, LangChain, LangGraph, CrewAI, Retrieval-Augmented Generation (RAG), AI Agents, Prompt Engineering, Model Fine-Tuning

Vector Databases Qdrant, FAISS, Pinecone, ChromaDB

Backend & Databases: FastAPI, SQLAlchemy, Redis, MySQL, PostgreSQL, SQLite

MLOps & Deployment: Docker, Kubernetes, MLflow, GitHub Actions, CI/CD, AWS SageMaker, Streamlit

Data Science & Statistics: Pandas, NumPy, Matplotlib, Plotly, Power BI, Descriptive Statistics, Inferential Statistics, A/B Testing, PCA, Regression Analysis, Probability Theory

Developer Tools: Git, GitHub, Jupyter Notebook, Google Colab, VS Code, PyCharm, Linux, Anaconda

Experience

Freelance AI/ML Engineer – Machine Learning, Deep Learning & Generative AI 2024 – Present

Independent AI Engineering & Consulting *Kathmandu, Nepal*

- Delivered end-to-end AI solutions for independent projects and client-oriented use cases across **Machine Learning, Deep Learning, Computer Vision, LLMs, RAG, and Multi-Agent AI**, independently handling architecture, development, evaluation, deployment, and documentation.
- Fine-tuned **Qwen3-4B-Instruct** using **QLoRA, PEFT, Unsloth, and TRL**, developing reproducible training and evaluation pipelines and exporting optimized model artifacts in **LoRA, FP16, and GGUF** formats.
- Engineered production-ready **RAG and AI backends** using **FastAPI, Qdrant, LangChain, Redis, and NVIDIA NIM**, implementing document ingestion, vector search, conversational memory, retrieval, and LLM-powered generation.
- Developed **multi-agent AI systems** with **CrewAI and LangGraph** for workflow automation, including hotel booking assistants, conversational agents, and compliance-document intelligence systems.
- Built and deployed **Computer Vision and medical imaging solutions** using deep learning and transfer learning, covering image classification, object detection, and medical image analysis with reproducible evaluation pipelines.
- Managed the complete software lifecycle from **problem definition and system architecture to implementation, testing, containerization, deployment, and technical documentation**, delivering reproducible projects through GitHub and Kaggle.

Machine Learning and Deep Learning Research Intern Aug 2025 – Apr 2026

NepAI Applied Mathematics and Informatics Institute for Research (NAAMII) – MAPMED Lab *Kathmandu, Nepal*

- Built end-to-end **medical image embedding extraction pipeline** processing 3D volumetric CT/MRI scans using MedicalNet ResNet models; implemented global and ROI-based embeddings across 4 feature layers (256–2048 dims).
- Engineered **3 fine-tuning strategies** (partial, full, linear probe) for automated esophagitis grading using the RTOG dataset, achieving **92%+ validation accuracy** via stratified k-fold cross-validation.
- Developed advanced **3D preprocessing pipeline** with center-crop normalization, multi-scale augmentation, and NIfTI format conversion, reducing image processing overhead by **65%** through batch automation.
- Created comprehensive **embedding analysis framework** with PCA dimensionality reduction, contrastive similarity metrics, Isolation Forest outlier detection, and 16+ automated visualizations for quality assurance.
- Designed robust data utilities including CSV label merging, DICOM/NIfTI validation, cross-platform logging, and production-ready error handling; documented **15+ technical workflows** for reproducible research.

Projects

Production-Ready Qwen3-4B LLM Fine-Tuning Pipeline | *Unsloth, Transformers, TRL, PEFT, QLoRA* Kaggle

- Designed and implemented an end-to-end **QLoRA fine-tuning pipeline** for **Qwen3-4B-Instruct** using **4-bit NF4 quantization**, PEFT, and Unsloth, enabling efficient supervised fine-tuning on dual NVIDIA T4 GPUs.
- Built a configurable data pipeline supporting both **human-curated** and **LLM-generated synthetic instruction datasets**, enabling reproducible experimentation across multiple training datasets.
- Optimized training with **TRL's SFTTrainer** by tuning LoRA rank, alpha, learning rate scheduling, gradient accumulation, sequence length, and mixed-precision settings for stable convergence under GPU memory constraints.

- Implemented automated experiment tracking by saving **training logs, checkpoints, evaluation metrics, generated outputs, tokenizer artifacts, and configuration files**, ensuring reproducibility and efficient model comparison.
- Developed an evaluation and inference pipeline to compare base and fine-tuned models through qualitative response analysis, validating instruction-following improvements before deployment.
- Exported trained models as **LoRA adapters, merged FP16 checkpoints, and GGUF** for deployment on Hugging Face Transformers, Ollama, and llama.cpp; published the complete notebook, model, and training artifacts on **Kaggle.**

AI-Powered Multi-Agent Hotel Booking Assistant | *Python, CrewAI, NVIDIA NIM, LiteLLM, JSONC* GitHub

- Developed an intelligent **multi-agent hotel booking system** using CrewAI, where specialized AI agents collaboratively handled travel consultation, hotel search, and booking assistance through sequential task orchestration.
- Designed autonomous agents to analyze customer travel preferences, including destination, budget, travel dates, hotel type, amenities, and special requests, enabling personalized hotel recommendations.
- Integrated **NVIDIA NIM Large Language Models** via LiteLLM to power context-aware reasoning, natural language understanding, and agent-to-agent collaboration throughout the booking workflow.
- Implemented context sharing and sequential execution between agents, allowing seamless information flow from requirement gathering to hotel recommendation and booking summary generation.
- Built a modular and extensible architecture supporting future integration with hotel search APIs, Google Maps, live pricing, payment gateways, RAG knowledge bases, and Streamlit-based user interfaces.
- Structured the project using configurable JSONC agent definitions, environment-based configuration, and reusable workflows to facilitate rapid customization and scalable deployment.

Neural RAG API Conversational AI | *FastAPI, LangChain, NVIDIA NIM, Qdrant, Redis, SQLAlchemy, Docker* GitHub

- Developed a production-ready **Retrieval-Augmented Generation (RAG)** backend enabling document-aware conversational AI with semantic retrieval, reducing LLM hallucinations by **up to 95%** through retrieval-grounded responses and context-aware generation.
- Built a custom RAG pipeline integrating **Qdrant Vector Database**, **NVIDIA embedding models**, and **LangChain**, delivering low-latency semantic search with citation-grounded responses.
- Implemented persistent multi-turn conversational memory using **Redis** with automatic session management and configurable TTL for enterprise-scale chat applications.
- Designed an intelligent document ingestion pipeline supporting PDF processing, semantic chunking, embedding generation, and scalable vector indexing for efficient knowledge retrieval.
- Engineered an LLM-powered interview booking extraction service that automatically extracts structured candidate information (name, email, interview date, and time) from natural language conversations into a SQL database.
- Containerized the application using **Docker Compose** and designed a modular service-oriented architecture with environment-based configuration for scalable production deployment.

ComplianceAI Multi-Agent Document Reconciliation System | *CrewAI, Python, Pydantic, Tesseract OCR* GitHub

- Designed and implemented an end-to-end **multi-agent document intelligence pipeline** using **CrewAI** to independently extract, cross-validate, and reconcile conflicting compliance data from manufacturer PDFs.
- Built independent source-specific extraction agents with **pdfplumber** text/table extraction and automatic **Tesseract OCR** fallback for scanned or unreliable PDF pages, preserving extraction method and source provenance.
- Developed a strongly-typed **Pydantic** extraction and reconciliation layer that classifies fields as **agree, conflict, document-only, or not-established**, while preserving both raw source values instead of silently selecting a winner.
- Implemented field-level normalization and cross-document reconciliation for product specifications, manufacturer information, test evidence, labeling, and importer documentation across multiple datasheet revisions and variants.
- Designed a **provenance-aware confidence framework** linking extracted values to their originating document, page, extraction method, and confidence level to improve traceability and auditability of compliance decisions.
- Generated structured **JSON reconciliation artifacts** and human-readable **Markdown compliance reports**, creating an end-to-end workflow from unstructured manufacturer documentation to review-ready compliance intelligence.

Education

National Infotech College, Tribhuvan University Nepal

Bachelor of Computer Application (BCA) – GPA: 3.34/4.00

Achievements & Certifications

AWS AI/ML Scholar 2025 – Selected among **top 3%** of applicants through competitive AWS AI & ML Scholarship Program based on technical excellence.

AWS Certified: Machine Learning Foundations – Hands-on expertise in ML principles, AWS services, and cloud-based model training.

AWS Certified: Generative AI with AWS – Practical experience with Amazon Bedrock, foundation models, and generative AI application development.

Positions of Responsibility

Teaching Assistant 2025

Annual Nepal AI School *Nepal*

- Supported instructors in delivering **AI and data science** sessions; assisted students with hands-on labs and doubt resolution, evaluated assignments, and coordinated workshops and technical events.